Gate-Controlled Hole Injection at Top and Edge Contacts to Few-Layer InSe: An Anisotropic, Electro-Thermal TCAD Study Verified Analytically and Benchmarked on a Known Contact

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Abstract

We model hole injection into back-gated, few-layer γ-InSe through top and edge metal contacts with an open-source DEVSIM model. The model couples Poisson's equation to Fermi–Dirac carriers with anisotropic (in-plane/out-of-plane) mobility. Contacts use flux-balanced thermionic emission with a van der Waals (vdW) gap transmission and nonlocal Tsu–Esaki tunnelling along the contact normals. The lattice heat equation has anisotropic conductivity and includes Joule, interface, Peltier and tunnelling heat. Nine verification tests reproduce closed-form results and conservation laws. As a numerical benchmark on a different material, with a single parameter fitted to the measured contact resistance of ultra-high-vacuum Au/MoS_2 contacts (740 $\Omega \cdot \mu$m), the same model gives a transfer length of 34 nm at 300 K (measured 35–40 nm) and the measured trends with temperature. For InSe we find: (i) With the Fermi-level pinning measured for direct metal/InSe contacts ($S = 0.05$, pinning level 0.3 eV below E_C) and the measured electron affinity, every common metal gives a hole barrier of 1.05–1.12 eV. Such contacts are electron-injecting, and hole conduction requires gate-induced tunnelling, which carries $\geq$99.9% of the on-current; the on-state resistance then varies by only 2.4$\times$ from Ti to Pt. (ii) For hole barriers up to 1 eV, edge contacts outperform 50-nm top contacts in the gated on state, and the advantage grows with the barrier: 14$\times$ at 0.2 eV, 51$\times$ at 0.6 eV and 150$\times$ at 1.0 eV. It arises from the vdW gap and from slow out-of-plane transport. (iii) Lowering the hole barrier needs weaker pinning: with the measured oxide interlayer ($S = 0.5$) Pt reaches 0.52 eV and a 700$\times$ lower edge resistance than with direct contacts. In ungated, barrier-limited operation the edge advantage disappears if edge contacts are more strongly pinned than top contacts; in the gated on state it survived independently varied pinning ($S = 0.05$–0.5) in 96% of Monte Carlo samples. (iv) Top contacts have an optimum film thickness of four to six layers, and edge contacts improve monotonically with thickness. (v) Edge-contacted devices carry more current and heat correspondingly more, but with directly pinned contacts self-heating is negligible and it becomes significant only after depinning; Peltier heat cools the source and heats the drain. A transmission-line compact model reproduces the simulated top-contact resistance with a median error of 11%, and Monte Carlo analysis identifies band alignment and pinning as the dominant uncertainties of the ungated device, and the pinning level and the hole tunnelling mass in the gated on state, where the edge contact has lower resistance in every sample with shared pinning (median ratio about 100).

Keywords: indium selenide; Schottky-barrier FET; contact resistance; tunnelling; electro-thermal simulation; van der Waals contacts; compact model; uncertainty quantification

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1 Introduction

InSe combines high electron mobility [1], ballistic short-channel transistors [2] and wafer-scale growth [3], but complementary logic also needs efficient hole injection, which has received far less attention [4]. Metal contacts to 2D semiconductors are governed by the Schottky barrier and Fermi-level pinning, by the vdW tunnel gap of top-deposited metals, and by how current spreads from the contact into the channel [5, 6]. Edge contacts remove the vdW gap and inject in-plane [7, 8]. Real devices are gated, and the gate can thin the Schottky barrier until tunnelling dominates injection, as in Schottky-barrier FETs. Layered crystals are also strongly anisotropic. Carrier mobility across InSe layers is typically one to three orders of magnitude lower than along them [9], and the through-plane thermal conductivity is about eleven times lower [10].

This paper asks which contact geometry gives the lowest resistance for hole injection into a few-layer InSe p-FET, and why. It also asks how robust that answer is to the interface parameters, and what the measured Fermi-level pinning of InSe contacts [11, 12] implies for p-type operation. We answer with a model in which each of the following ingredients is included and tested:

- a flux-balanced contact condition;
- non-equilibrium Poisson coupling;
- Fermi–Dirac statistics;
- a gate;
- nonlocal tunnelling;
- transport and thermal anisotropy;
- a complete heat source.

We also benchmark its numerics on a well-characterised metal/2D contact (Au/MoS_2) and test its predictions out of sample.

2 Model

2.1 Device and band structure

The device is an N-layer InSe film (0.83 nm per layer; $N = 6$, $t = 5$ nm by default) on 10 nm SiO_2 with a back gate (flat band at $V_G = 0$), with a 50-nm channel. Top contacts of length L_c cover the upper surface, and edge contacts cover the film's end faces. The band gap follows a fit to photoluminescence data [13, 14], $E_g(N) = 1.26 + 2.78N^{-1.70}$ eV, which gives 1.69 eV at 3 layers, 1.39 eV at 6 layers and 1.30 eV at 12 layers. 32% of the confinement shift is assigned to the valence band, roughly $m_e/(m_e + m_h)$. The 6-layer film therefore has $E_V = 5.00$ eV below vacuum (Fig. 10a). All parameters and their provenance are listed in Table 1. The two geometries are sketched in Fig. 1. Sections 3.2–3.10 use this *reference alignment* with $S = 1$, which isolates the effects of geometry, transport and tunnelling at a given barrier. Section 3.5 then uses the measured electron affinity $\chi = 4.6$ eV of bulk InSe [11], which places E_V of the 6-layer film at 5.90 eV, together with the measured and calculated pinning of InSe contacts. Calculated ionisation energies are lower, about 5.2 eV [12], so the absolute alignment is uncertain by several tenths of an eV; results are therefore also given versus the hole barrier ϕ_{Bp}, which is the quantity that matters.

Table 1: Parameters (6-layer InSe unless stated). M: measured; L: literature calculation or compilation; R: representative value within a measured range; A: assumed. The last column gives the effect of a larger value on the on-state resistance (or on $\Delta T_{\max}$), from the Monte Carlo analysis (Sec. 3.10, Fig. 16) and the sensitivity runs, and the expected change for thinner films where it is known.

Parameter	Value	Source / type	Effect of a larger value; thickness trend
$E_g(N)$, $E_V(N)$	$1.26 + 2.78N^{-1.7}$ eV; 1.39, 5.00 eV at $N = 6$	fit to PL [13, 14] (M); 32% VB share (A)	Deeper E_V raises ϕ_{Bp} and R exponentially. Thinner films: larger gap, higher R (Fig. 10); VB share 0.2–0.5 moves the top-contact optimum by one layer
m_e, m_h (DOS and tunnelling)	0.07, 0.17 m_0	L [15]	Heavier m_h suppresses tunnelling and raises the gated R (second-largest index). Few-layer value uncertain; direction unknown
ε_r	6.4	L	Raises the gated R moderately ($\rho \approx$ 0.2–0.3). Weaker screening in few layers would lower R
$\mu_{n,\parallel}$, $\mu_{p,\parallel}$	1000, 50 cm^2/Vs	M [1]; A	Lowers only the channel part; contact-limited on state insensitive (index at noise floor). Thinner films: lower μ, higher R_{ch}
$\mu_\perp/\mu_\parallel$	0.01 (range 0.002–1)	M (electrons; 0.002 in vertical FETs [9], ∼0.01 bulk)	Smaller ratio raises top-contact R (2× from 1 to 0.002, ungated); edge unaffected
$\mu(T)$	$\propto T^{-1.5}$	A	Stronger decay raises R of self-heated devices only
$\kappa_\parallel$, $\kappa_\perp$	8.5, 0.76 W/mK	M, time-domain thermoreflectance [10]	Lower κ raises $\Delta T_{\max}$ (measured vs isotropic $\kappa_\perp$: +8–17%). Few-layer κ likely below bulk: higher ΔT
c_V	1.43 J/cm^{3}K	Dulong–Petit (L)	Scales only the thermal time constant
N_A	10^{16} cm^{-3}	A	Weak (index at noise floor)
S_{top}, S_{edge}, Φ_{CNL}	1, 1, 4.35 eV (reference); 0.05 and 0.5 measured, 0.18 calculated (Sec. 3.5)	A (varied); M [11]; L [12]	Dominant: the pinning level and S set ϕ_{Bp} (largest indices); for Pt a larger S lowers R. $S_{\mathrm{edge}} > S_{\mathrm{top}}$ reverses the ungated ranking only
vdW gap d, Φ_{vdW}	0.30 nm, 2.0 eV ($\mathcal{T}_{\mathrm{vdW}} =$ 0.013)	A (varied)	Raise top-contact R only ($\rho \approx$ 0.2 gated); edge unaffected
oxide; R_{TBR}	10 nm SiO_2 (κ=1.4 W/mK); 10^{-8} m^{2}K/W at the gate	A	3× larger R_{TBR} raises $\Delta T_{\max}$ by 58–73%. Oxide thickness not varied

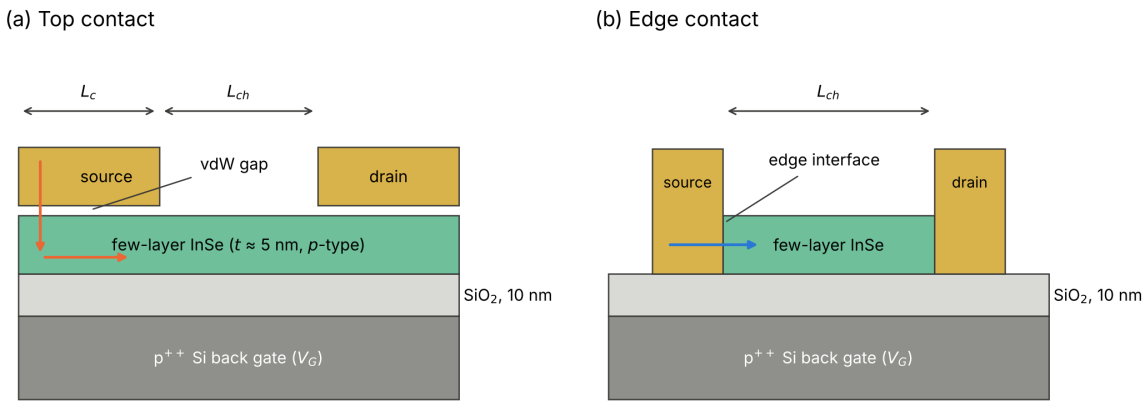


Figure 1: Simulated geometries (not to scale). (a) Top contacts of length L_c separated from the film by a van der Waals gap; holes cross the gap and then flow laterally. (b) Edge contacts abutting the end faces of the film. Both devices sit on 10 nm SiO_2 with a back gate.

2.2 Carrier transport

Poisson's equation with $\rho = q(p - n + N_D - N_A)$ is solved in InSe and oxide together with Scharfetter–Gummel continuity equations. Fermi–Dirac statistics enter through a degeneracy correction $\Delta(u) = \eta(u) - \ln u$ ($u = p/N_V$; Nilsson inverse, accuracy better than $0.006kT$) added to the SG drift potential, and through $p_{\mathrm{eq}} = N_V F_{1/2}(-\phi_{Bp}/kT)$ at the contacts [16]. Barriers with $\Phi_{\mathrm{eff}} > E_V$ are therefore allowed (degenerate contacts). Mobility is the tensor $\mu(\hat{\mathbf{e}}) = \mu_{\parallel}(\hat{e}_x^2 + r_\mu \hat{e}_y^2)$ along each mesh edge, which is exact on the axis-aligned mesh used. Thermal velocities, N_C, N_V, n_i and the contact injection follow the local lattice temperature.

2.3 Contacts and tunnelling

The effective work function is $\Phi_{\mathrm{eff}} = \Phi_{\mathrm{CNL}} + S(\Phi_M - \Phi_{\mathrm{CNL}})$, with separate S_{top} and S_{edge}. Carriers at the interface obey the flux condition $J_{p,s} = q v_R \mathcal{T}_{\mathrm{vdW}}(p_s - p_{\mathrm{eq}})$, with $v_R = A^* T^2/qN_V$, balanced against the drift-diffusion current; this is the thermionic-diffusion contact of Crowell and Sze [17]. Tunnelling below the barrier top is treated nonlocally, in the spirit of Ieong *et al.* [18]. For every contact node, the band edge along the inward normal defines the WKB transmission $\mathcal{T}(E)$ to the classical turning point. The Tsu–Esaki current density is

$$J(E)\,dE = \frac{A^* T}{k}\,\mathcal{T}_{\mathrm{vdW}}\,\mathcal{T}(E)\ln\frac{1 + e^{(\epsilon_{F,m} - E)/kT}}{1 + e^{(\epsilon_{F,s}(x_t) - E)/kT}}\,dE, \tag{1}$$

and it is deposited as generation at the turning point. The generation is written semi-implicitly as $G = a - b\,p$ (metal supply minus semiconductor back-flow), so it vanishes identically at equilibrium. It is iterated with under-relaxation until carrier densities change by less than 10^{-3}. Top-contact paths are vertical through the film and also include $\mathcal{T}_{\mathrm{vdW}}$. For comparison, a local field-enhancement model $1 + \Gamma(F)$ of the interface velocity [19] and a no-tunnelling model are also run.

2.4 Heat

The lattice heat equation $c_V \partial_t T = \nabla \cdot (\boldsymbol{\kappa} \nabla T) + H$ has $\boldsymbol{\kappa} = \mathrm{diag}(\kappa_{\parallel}, \kappa_{\perp})$ and

$$H = -\mathbf{J}_p \cdot \nabla \phi_p - \mathbf{J}_n \cdot \nabla \phi_n - T\mathbf{J}_p \cdot \nabla P_p - T\mathbf{J}_n \cdot \nabla P_n, \qquad P_p = \tfrac{k}{q}(2 - \eta_p), \tag{2}$$

which contains Joule heat and Peltier–Thomson heat [20]. Each contact additionally receives the interface Joule heat $J_s \Delta\phi$ and the junction Peltier heat $J_s T P_s$. Tunnelling carriers deposit their energy relative to the local mean carrier energy at the turning point, and the metal side receives the complementary Nottingham-type term. With these terms the total heat equals IV (Sec. 2.6). Heat leaves through the oxide to the gate, which is a heat sink with a thermal boundary resistance R_{TBR}. Metal contacts are adiabatic unless stated otherwise.

2.5 Extraction

$R_{\mathrm{tot}} = V_D / I_D$ is reported per unit width; 1 $\Omega \cdot$ cm equals 10^4 $\Omega \cdot \mu$m. It is decomposed exactly into $R_{c,s} + R_{\mathrm{ch}} + R_{c,d}$ using the majority-carrier quasi-Fermi potential on the film centre line at the inner contact edges.

2.6 Verification

Table 2 and Fig. 2 summarise nine tests: (V1) an ohmic resistor; (V2) a finite-velocity contact resistor; (V3) Mott–Gurney space-charge-limited current with Boltzmann and Fermi–Dirac statistics [21]; (V4) thermionic emission at 300 and 350 K; (V5) energy and charge conservation in 2D, gated and ungated (heat out equals IV within 0.3% except 1.2% in the hottest gated case, where Thomson heat from temperature gradients is not separately balanced); (V6) nonlocal tunnelling against an independent quadrature on the same band profile (0.17%), and on a depletion-approximation profile (13%, the approximation's own error); (V7) a degenerate contact at equilibrium; (V8) transient energy conservation and approach to the DC state; and (V9) Gauss's law with a gate. R_{tot} changes by less than 1.3% between 0.5× and 3× mesh density in the ungated device (Supplementary), and by at most 1.5% (with $\Delta T_{\max}$ changing by at most 0.6 K) at the operating points of Table 3 between 1× and 3×. Convergence is slower in the tunnelling-dominated on state at the measured pinning (Sec. 3.5): between 1× and 3× the edge-contact resistance changes non-monotonically by up to 12% and the top-contact resistance by up to 5%. On-state resistances and top/edge ratios there carry a mesh uncertainty of about 15% (Fig. 8c).

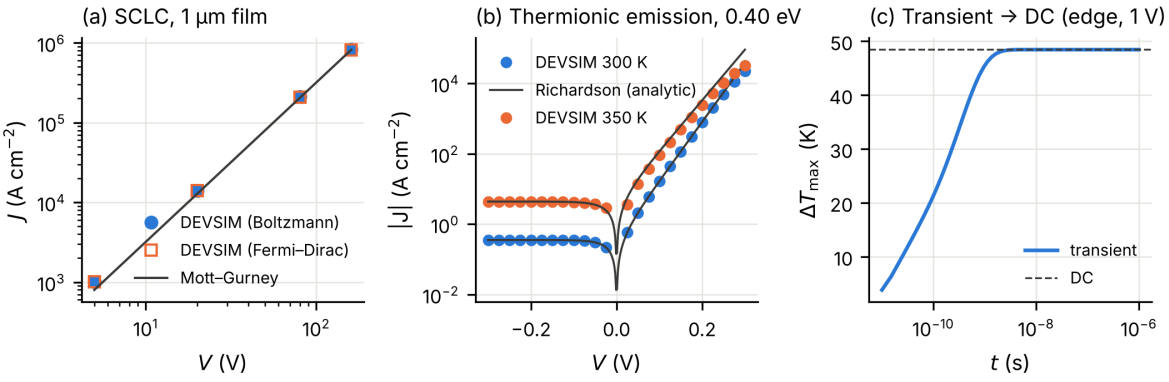


Figure 2: Verification. (a) Mott–Gurney law with Boltzmann and Fermi–Dirac statistics. (b) Thermionic emission at 300 and 350 K. (c) Transient heating of a gated edge device stepped to 1 V (tunnelling off), approaching the DC solution.

Table 2: Verification summary (`data_v3/verification_summary.csv`).

Test	Quantity	Relative error
V1	R ohmic (Ohm cm^2)	+1.9e-14
V2	R Robin v=1e+09 (Ohm cm^2)	+4.5e-10
V2	R Robin v=1e+05 (Ohm cm^2)	+4.5e-05
V2	R Robin v=1e+04 (Ohm cm^2)	+1.2e-03
V3	boltzmann J/J_MG at 160 V	+7.5e-03
V3	fd J/J_MG at 160 V	-5.5e-03
V4	ideality (300 K)	-6.3e-03
V4	|J_rev(0.3V)| (300 K)	-1.2e-03
V4	ideality (350 K)	-9.4e-03
V4	|J_rev(0.3V)| (350 K)	-1.4e-03
V5	top 4.8eV 0.05V P_out/IV	-2.7e-04
V5	top 4.8eV 0.05V -I_src/I_drn	+1.3e-14
V5	top 5.0eV 0.5V P_out/IV	+2.7e-04
V5	top 5.0eV 0.5V -I_src/I_drn	-2.8e-14
V5	top-gated 4.8eV 0.05V P_out/IV	-4.7e-04
V5	top-gated 4.8eV 0.05V -I_src/I_drn	+0.0e+00
V5	top-gated 5.0eV 0.5V P_out/IV	+2.3e-03
V5	top-gated 5.0eV 0.5V -I_src/I_drn	+5.4e-15
V5	edge 4.8eV 0.05V P_out/IV	+1.4e-03
V5	edge 4.8eV 0.05V -I_src/I_drn	-1.8e-14
V5	edge 5.0eV 0.5V P_out/IV	+2.9e-03
V5	edge 5.0eV 0.5V -I_src/I_drn	+4.4e-16
V5	edge-gated 4.8eV 0.05V P_out/IV	+7.5e-04
V5	edge-gated 4.8eV 0.05V -I_src/I_drn	-1.8e-16
V5	edge-gated 5.0eV 0.5V P_out/IV	+1.2e-02
V5	edge-gated 5.0eV 0.5V -I_src/I_drn	+2.0e-15
V6	J_tun: model vs quad on same profile	-1.7e-03
V6	J_tun: model vs depletion-approximation quadrature (approximation difference)	-1.3e-01
V7	p_s / (N_V F1/2(-phiB/kT)) [exact F]	-4.7e-04
V7	|I(V=0)| (A/cm^2)	4.8e-07 (abs.)
V7	max|phi_p| at equilibrium (V)	1.7e-04 (abs.)
V8	(E_out + E_stored)/E_in (0-1 us)	-4.2e-04
V8	T_max(1 us) - 300 vs DC	+0.0e+00
V9	(sum contact charge + semi charge)/|Q_gate|	-2.5e-06

2.7 Numerical benchmark on Au/MoS_2

No measurements of hole injection into p-InSe exist that could fix the interface parameters. We therefore use a different, well-characterised system only as a numerical benchmark of the solver, contact and tunnelling implementation: the best-characterised metal/2D contact, ultra-high-vacuum Au on 4.5-nm MoS_2 with a 90-nm SiO_2 back gate [22]. The measured quantities at $n = 10^{13}$ cm^{-2} are $R_c = 740\ \Omega\cdot\mu$m, $\rho_c \approx 3\times10^{-7}\ \Omega\,$cm^2 and $L_T \approx 35$–40 nm at 300 K, and $\rho_c \approx 1.5\times10^{-7}\ \Omega\,$cm^2 and $L_T \approx 20$ nm at 400 K. We keep the 0.15-eV electron barrier assumed in that work and use literature MoS_2 parameters, with an under-contact mobility of 35 cm^2/Vs scaling as $T^{-1.6}$. We fit a single parameter, $\mathcal{T}_{\mathrm{vdW}}$, to $R_c(300\,\mathrm{K})$ only. The quantities predicted out of sample are then $L_T(300\,\mathrm{K})$ (from simulated $R_c(L_c)$ with $L_c = 20$–200 nm), and ρ_c and L_T at 400 K.

3 Results

3.1 Benchmark and out-of-sample test

The fit gives $\mathcal{T}_{\mathrm{vdW}} = 1.0$, at its physical bound: $R_c = 731\ \Omega\cdot\mu$m against 740 $\Omega\cdot\mu$m measured. With the literature barrier, nonlocal tunnelling carries 99.8% of the injection, and no additional vdW tunnel barrier is needed. This is consistent with the clean interfaces of that study. Without further adjustment the model gives $L_T = 34$ nm and $\rho_c = 2.7\times10^{-7}\ \Omega\,$cm^2 at 300 K: ρ_c lies within the measured range, and L_T is 3% below the lower end of the measured 35–40 nm. At 400 K it predicts $\rho_c = 2.0\times10^{-7}\ \Omega\,$cm^2 (measured 1.5) and $L_T = 29$ nm (measured ≈20, quoted uncertainty up to 50%), with the measured direction of change for both (Fig. 3). The model underestimates how much L_T shrinks with temperature. This is consistent with its temperature-independent barrier and no phonon-assisted injection. We take the agreement as validation of the contact and tunnelling implementation at the 10–40% level. It is a check of the numerics only: MoS_2 has a different band structure, no strong anisotropy and no layer-dependent gap, so it does not validate the InSe parameters or the InSe-specific results.

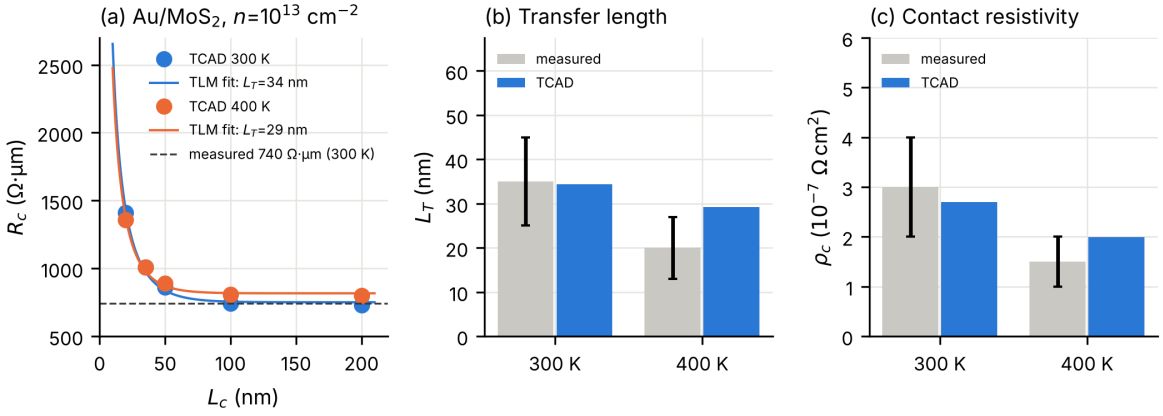


Figure 3: (a) Simulated $R_c(L_c)$ of Au/MoS_2 at 300 and 400 K with transmission-line fits. (b) Transfer length and (c) specific contact resistivity against the measurements of Ref. [22], with their quoted uncertainties. Only R_c at 300 K was fitted; the 400 K values and L_T are predictions.

3.2 Ungated InSe: work function, statistics and tunnelling

Without a gate (Fig. 4), R_{tot} at 0.05 V falls exponentially with Φ_M down to $E_V = 5.0$ eV. For 4.8 eV it is 1430 $\Omega \cdot$ cm (edge), 5080 $\Omega \cdot$ cm (top, 50 nm) and 11 500 $\Omega \cdot$ cm (top, 10 nm). With Fermi–Dirac statistics the resistance keeps decreasing for degenerate contacts: at 5.65 eV it is 5.4, 7.3 and 7.7 $\Omega \cdot$ cm, a factor 2.1–2.6 below the Boltzmann result, which is clipped at E_V. Contacts carry 67% (top, 50 nm) and 87% (top, 10 nm) of R_{tot} at 4.8 eV, against 7.5% for the edge. In the ungated, fully depleted film, nonlocal tunnelling carries 44% of the edge-contact current and lowers its resistance by 13% relative to pure thermionic emission. For top contacts it carries $\leq$8.5% and lowers resistance by 2–6%. For the edge contact the local $\Gamma(F)$ model recovers only a fifth of this reduction (2.3% vs 12.6%), because only nonlocal injection bypasses the diffusion-limited part of the barrier; for top contacts, where injection is limited by the interface, the two models differ by about 1%.

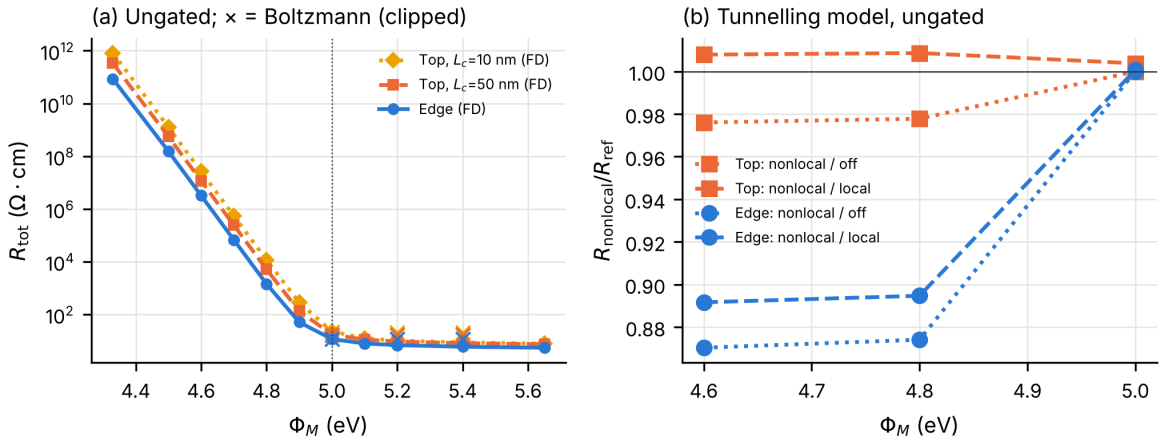


Figure 4: Ungated devices. (a) R_{tot} versus Φ_M with Fermi–Dirac statistics (lines) and Boltzmann statistics clipped at E_V (×). (b) Ratio of nonlocal-tunnelling resistance to the no-tunnelling and local-Γ models.

3.3 Anisotropy and top-contact length

Out-of-plane transport limits top contacts (Fig. 5). Lowering $\mu_\perp/\mu_\parallel$ from 1 to 0.002 doubles the top-contact resistance at 4.8 eV (3920→7990 $\Omega \cdot$ cm for 50 nm; 8270→16 700 $\Omega \cdot$ cm for 10 nm), whereas edge contacts are unaffected. The edge advantage at 4.8 eV therefore grows from 64% (isotropic) to 72% (baseline) and 82% ($r_\mu = 0.002$). The top-contact resistance saturates for $L_c \gtrsim$ 50–100 nm, which indicates transfer lengths of a few tens of nanometres. Thermal anisotropy matters little at this thickness: with $\kappa_\perp = 0.76$ instead of 8.5 W/mK, the temperature rise of ungated devices at 1 V changes by less than 4% (the gated operating point of Sec. 3.8 is more sensitive: 8–17%).

3.4 Pinning: can top contacts win?

Edge contacts expose dangling bonds and may be more strongly pinned than vdW top contacts. Figure 6 varies S_{edge} with $S_{\mathrm{top}} = 1$. Already at $S_{\mathrm{edge}} = 0.7$ the edge contact is 180× more resistive than with $S_{\mathrm{edge}} = 1$ at 4.8 eV (2.6×10^5 $\Omega \cdot$ cm), and 50× worse than the 50-nm top contact. It overtakes the top contact again only for metals with $\Phi_M \geq 5.4$ eV. The geometric and vdW

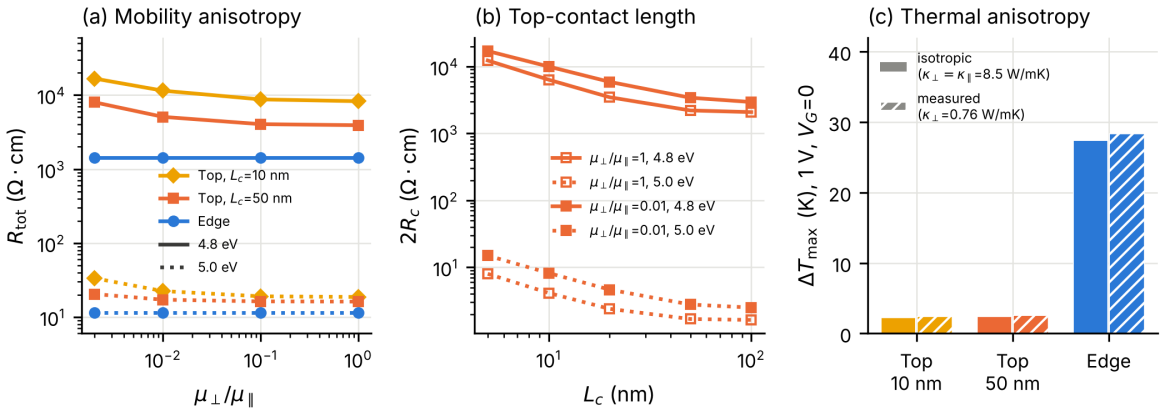


Figure 5: (a) R_{tot} versus out-of-plane/in-plane mobility ratio (colour: geometry; line style: Φ_M). (b) $2R_c$ of top contacts versus contact length for two mobility ratios. (c) Peak temperature rise of ungated devices at 1 V for isotropic and measured $\kappa_\perp$.

advantage of edge contacts is therefore smaller than the effect of a modest difference in pinning. Which geometry wins is decided by interface chemistry, and S_{edge} is the quantity experiments should measure first. This reversal applies to the ungated, barrier-limited contact. In the gated on state the edge contact keeps its advantage over a wide range of unequal pinning (Sec. 3.10).

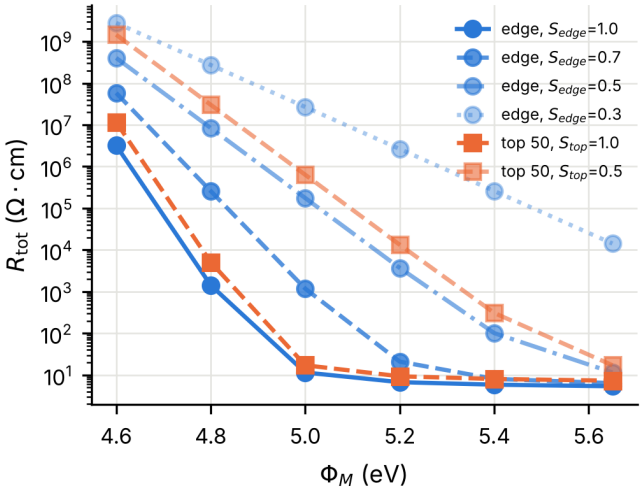


Figure 6: R_{tot} of edge contacts for $S_{\mathrm{edge}} = 1$–0.3 and of 50-nm top contacts for $S_{\mathrm{top}} = 1$ and 0.5 (ungated, 0.05 V).

3.5 Measured pinning and literature band alignment

Fermi-level pinning has been measured for InSe. Chen *et al.* [11] extracted barriers for In, Ti, Cr and Pd contacts and found $S = 0.05 \pm 0.02$ for direct metal/InSe contacts, with the pinning level about 0.3 eV below E_C; a one-layer oxide interlayer raised S to 0.5. Hybrid-functional calculations for ideal, defect-free interfaces give $S = 0.18$ with metal-induced gap states pinning E_F about

0.58 eV above E_V [12]. We simulated Ti, Cr, Au, Pd and Pt (4.33–5.65 eV) for these three cases and for $S = 1$, with $\chi = 4.6$ eV and $S_{\text{top}} = S_{\text{edge}}$, and swept the hole barrier directly from 0 to 1 eV (Fig. 7). The on state is $V_G = -3$ V at $V_D = 0.05$ V.

With the measured direct-contact pinning every metal gives $\phi_{Bp} = 1.05$–1.12 eV and an electron barrier of only about 0.3 eV. Direct metal contacts to InSe are therefore electron-injecting, which agrees with the n-type behaviour of most InSe transistors, and an ungated device conducts electrons rather than holes. Under a negative gate, holes are injected almost entirely by tunnelling ($\geq$99.9%), and the edge on-state resistance is 5.2×10^3 $\Omega \cdot$cm for Pt and 1.2×10^4 $\Omega \cdot$cm for Ti, only 2.4× apart for a 1.3-eV range of work function. The choice of metal hardly matters.

With the calculated ideal pinning the barriers fall to 0.52–0.76 eV and the edge on-state resistance to 7.1 $\Omega \cdot$cm (Pt) to 123 $\Omega \cdot$cm (Ti). The measured oxide interlayer gives the same 0.52 eV for Pt, a 700× improvement over the direct Pt contact, but 0.79 eV for Pd and Au. Only an unpinned Pt contact reaches 0.25 eV and $R_{\text{tot}} = 0.51$ $\Omega \cdot$cm. Lowering the barrier, by depinning the interface or by choosing a high-work-function metal once it is depinned, is therefore the first lever for p-type InSe; geometry is the second.

The geometry lever becomes stronger, not weaker, as the barrier rises (Fig. 7b). The ratio of top- to edge-contact on-state resistance is 4.1 at $\phi_{Bp} = 0$, 14 at 0.2 eV, 35 at 0.4 eV, 51 at 0.6 eV, 74 at 0.8 eV and 150 at 1.0 eV. At high barriers the current must tunnel, and the edge contact tunnels in-plane with the gate field, while the top contact must tunnel vertically through the vdW gap and the slow out-of-plane path. Without tunnelling the resistance rises by 5×10^2 (edge) and 3×10^2 (top) at 0.2 eV and by 5×10^7 and 8×10^6 at 0.6 eV, so a thermionic contact model would declare any InSe hole contact unusable. Gated runs that did not converge are omitted. The hole current there is expected to be small, but this was not verified. They are the top contacts with $\phi_{Bp} \geq 1.07$ eV (direct pinning except Pt; Ti and Cr with $S \geq 0.5$) and the runs without tunnelling at 0.8–1.0 eV. For the measured direct pinning the top/edge ratio is therefore available only for Pt (170× at $\phi_{Bp} = 1.05$ eV). The ungated resistance is not monotonic above $\phi_{Bp} \approx 0.7$ eV because electron injection takes over.

As a consistency check against the measured electron barriers, we applied the same model to electron injection into an n-type film ($N_D = 10^{16}$ cm^{-3}) with the measured direct pinning, for In, Ti and Pd contacts at 200–400 K (Fig. 8a,b). At $V_G = 0$ the Arrhenius barriers, from the slope of $\ln(I/T^2)$ against $1/T$, are 0.14–0.17 eV (edge) and 0.20–0.24 eV (top). They differ by at most 0.04 eV between metals across a 1-eV range of work function, in line with the weak metal dependence reported in Ref. [11]. They lie 0.06–0.16 eV below the input barrier of 0.30 eV because tunnelling contributes even without a gate, so barriers extracted this way are lower bounds. Above $V_G = +1$ V the Arrhenius slope turns negative because channel transport dominates.

3.6 Gated Schottky-barrier p-FETs

With the gate (Fig. 9), both geometries switch with on/off ratios above 10^{12} between $V_G = +1$ and -4 V. At $\Phi_M = 4.8$ eV the on-current at $V_D = 0.05$ V and $V_G = -4$ V is 0.21 A/cm (edge) and 0.014 A/cm (top, 50 nm). Practically all of it is injected by tunnelling through the gate-thinned barrier: the no-tunnelling model gives 840× (edge) and 450× (top) less current, and the local $\Gamma(F)$ model 60× and 55× less. At 5.0 eV tunnelling still carries 80–83% of the on-current; at 5.2 eV, where the contacts are degenerate, only 1–16%. At 5.0 eV and $V_G = -4$ V the contact resistance is $R_c = 86$ $\Omega \cdot \mu$m (edge) against 2080 $\Omega \cdot \mu$m (top), a factor of 24, while the channel resistances are nearly equal. The contact therefore limits the top-contacted transistor, and the channel limits the edge-contacted one.

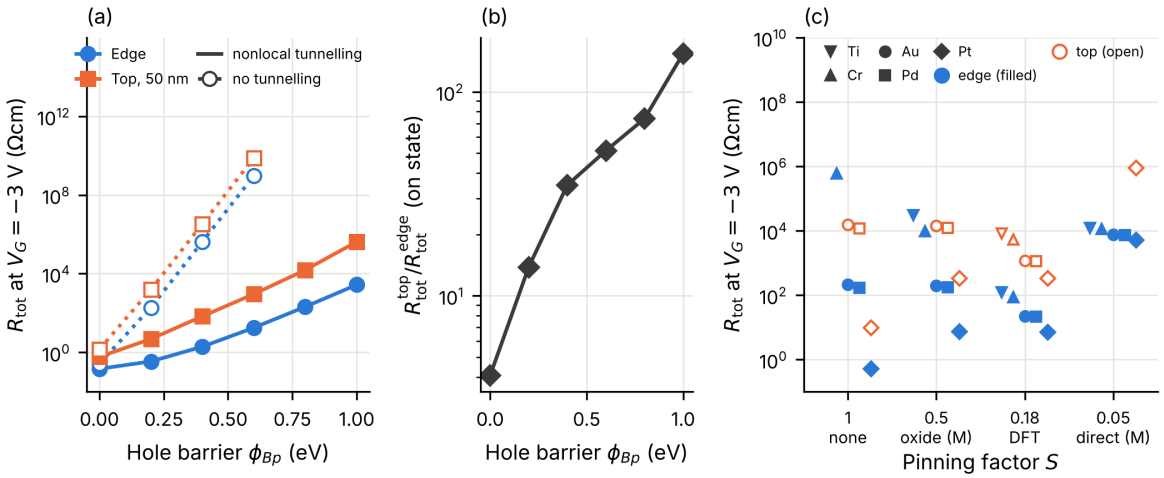


Figure 7: Literature band alignment ($\chi = 4.6$ eV, $E_V = 5.90$ eV for 6 layers). (a) On-state R_{tot} ($V_G = -3$ V, $V_D = 0.05$ V) versus hole barrier, with nonlocal tunnelling (solid) and without tunnelling (dotted). (b) Ratio of top- to edge-contact on-state resistance. (c) On-state R_{tot} for five metals and four pinning cases: no pinning; measured oxide interlayer and direct contact (M) [11]; calculated ideal interface [12]. Filled: edge; open: top, 50 nm. Missing points did not converge and were not determined.

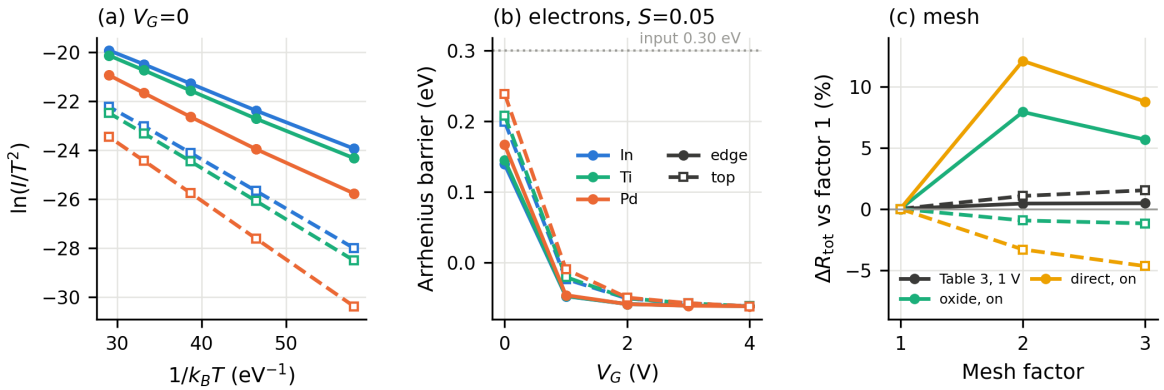


Figure 8: (a) Arrhenius plots of electron injection at $V_G = 0$ with the measured direct pinning ($S = 0.05$, input barrier 0.30 eV) for In, Ti and Pd; filled: edge, open: top. (b) Arrhenius barrier versus V_G. (c) Change of R_{tot} with mesh factor for the Table 3 operating point and for the gated on state at the measured pinning (Pt, $V_G = -3$ V).

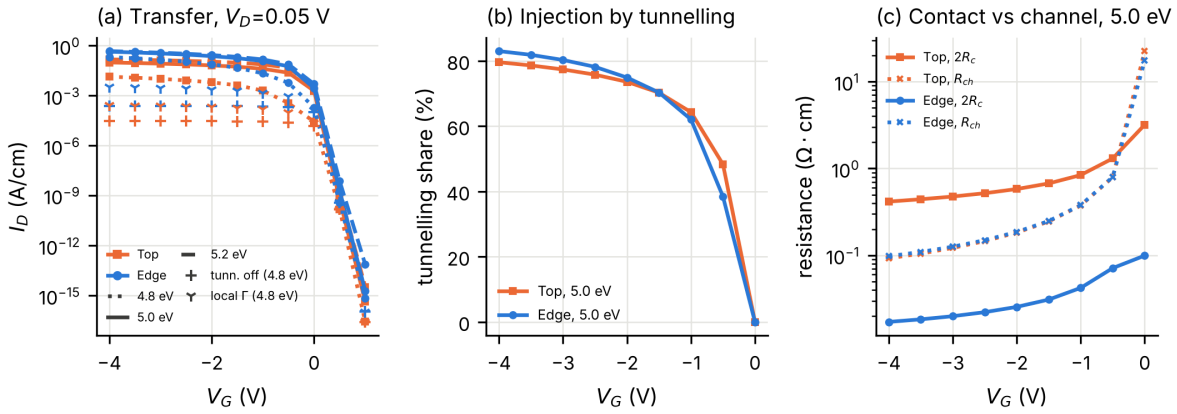


Figure 9: Gated devices (10 nm SiO_2, $V_D = 0.05$ V). (a) Transfer curves for the full nonlocal model (lines), with the no-tunnelling (+) and local-Γ (Y-shaped symbols) models at 4.8 eV. (b) Share of the current injected by tunnelling. (c) Contact and channel resistance at 5.0 eV.

3.7 Thickness

The trends below are robust to the band alignment and to the assumed 32% valence-band share of the confinement shift (Fig. 12b,c). With the measured alignment and Pt contacts ($V_G = -3$ V) edge contacts again improve monotonically from 3 to 12 layers (oxide interlayer: 25→4.5 Ω · cm), and top contacts have an optimum at five layers (interlayer, 207 Ω · cm) or six layers (direct pinning). Changing the valence-band share to 0.2 or 0.5 moves the optimum of the reference alignment between four and five layers. Thinner films have larger band gaps and deeper valence bands (Fig. 10a), so ungated resistances rise steeply below six layers. At 4.8 eV the edge-contact resistance rises from 242 Ω · cm (12 layers) to 1.1×10^5 Ω · cm (3 layers). In the gated on state ($\Phi_M = 5.0$ eV, $V_G = -3$ V) the edge contact improves monotonically with thickness (0.197→0.135 Ω · cm from 3 to 12 layers). The top contact instead has an optimum near four layers: 0.53 Ω · cm, rising to 1.55 Ω · cm at 12 layers. Beyond that thickness the vertical access through layers with low $\mu_\perp$ outweighs the shallower barrier (Fig. 10b).

3.8 Self-heating

Table 3 compares the geometries at the same gated operating point ($\Phi_M = 5.0$ eV, $V_G = -2$ V). At $V_D = 1$ V the edge-contacted device carries 3.7 A/cm and heats by 150 K; the 50-nm top contact carries 1.16 A/cm and heats by 24 K. Per unit power the edge device is also about twice as hot (41 vs 21 K cm/W), because its footprint on the oxide is a third of that of the top-contacted film. In the on state at $V_G = -3$ V (Fig. 11a) the edge device reaches about 490 K at 1 V, so its usable bias is thermally limited, not contact-limited. The two-dimensional maps in Fig. 13 show where the heat sits: in the top-contacted device the hottest point lies at the drain-side channel end on the contact side of the film, and the whole film stays within 24 K of ambient; in the edge-contacted device the film is 126–150 K hot everywhere and is coolest at the oxide interface, through which heat leaves to the substrate.

Table 3 uses the $S = 1$ reference alignment, where $\Phi_M = 5.0$ eV gives a hole barrier near zero. At the measured alignment (Pt, $V_G = -3$ V, $V_D = 1$ V; Fig. 12a) self-heating depends on the pinning. Directly pinned contacts carry so little current that the edge device heats by only 0.2 K. With the oxide interlayer or the calculated ideal pinning ($\phi_{Bp} \approx 0.52$ eV) the edge device heats

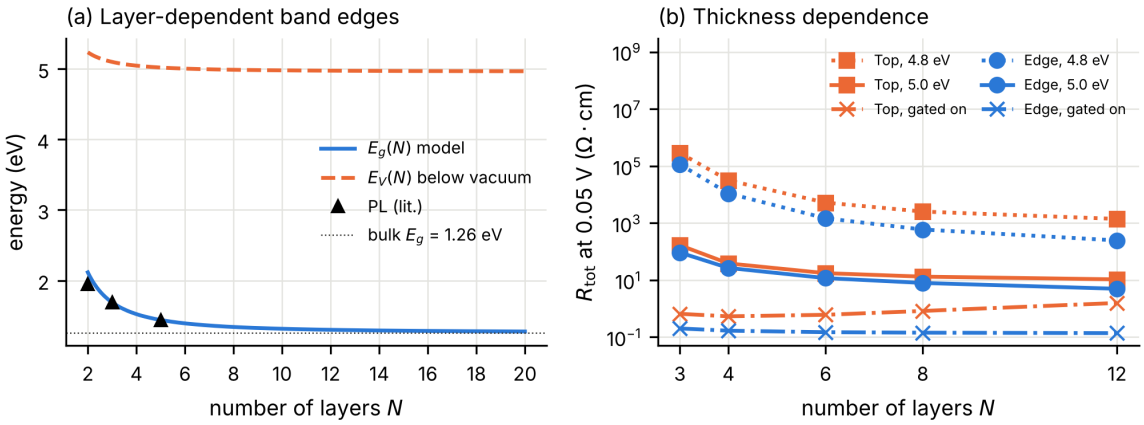


Figure 10: (a) Layer-dependent band gap and valence-band edge with photoluminescence data. (b) R_{tot} versus layer number, ungated and in the gated on state ($V_G = -3$ V, $\Phi_M = 5.0$ eV).

by 24 K and the top-contacted one by 0.2 K. An unpinned Pt contact ($\phi_{Bp} = 0.25$ eV) heats the edge device by 128 K (top: 4.4 K). Self-heating is therefore a concern only once the contacts are depinned. A three times larger R_{TBR} (3×10^{-8} m^{2}K/W) raises these rises by 58–73%, and heat sinking through the metal lowers them by 22–23%. Metal contacts are adiabatic in the baseline model, so the rises quoted are upper bounds. At the operating points of Table 3 the heat leaving the device equals IV within 0.5% (top) and 1.2–1.9% (edge, 0.5 and 1 V). For the 150 K edge rise this is about 3 K, larger than the Peltier effect below, so temperatures are quoted to ±3 K.

The individual heat terms matter at the 7–42% level (Fig. 11b):

- Using bulk Joule heat only underestimates ΔT_{max} by 7–8% and misses 8–15% of the dissipated power.
- Measured out-of-plane conductivity, compared with isotropic κ, raises ΔT_{max} by 8% (edge) and 17% (top).
- Heat sinking through the metal contacts lowers ΔT_{max} by 23% (edge) and 42% (top).
- A ten times smaller thermal boundary resistance lowers it by 35–37%.

Peltier heat changes ΔT_{max} by only about 1 K but redistributes heat: it cools the hole-injecting contact and heats the collecting one (top, 1 V: source +11.9 K, drain +21.6 K with Peltier, against +13.5 and +20.9 K without). Temperature-dependent statistics change temperatures by less than 1%. The lattice responds to a drain step within a few hundred picoseconds, with 63% of the final rise at 0.20 ns (top) and 0.25 ns (edge) (Fig. 11c; this transient was simulated without tunnelling, so its amplitudes are smaller than in Table 3). For pulses longer than a nanosecond the DC temperatures therefore apply.

3.9 Compact model

For top contacts in the barrier-limited regime we compare a transmission-line model with the simulation. The model uses

- an interface resistivity $\rho_{\mathrm{if}} = V_t/(q v_R \mathcal{T}_{\mathrm{vdW}} p_{\mathrm{eq}})$;

Table 3: Gated operating points ($\Phi_M = 5.0$ eV, $V_G = -2$ V, baseline thermal model). All quantities in a row refer to the same bias.

Geometry	V_D (V)	I_D (A/cm)	IV (W/cm)	$\Delta T_{\max}$ (K)	ΔT_{source} (K)	ΔT_{drain} (K)
Top, 50 nm	0.5	0.618	0.309	5.9	3.5	5.6
Top, 50 nm	1.0	1.164	1.164	24.0	11.9	21.6
Edge	0.5	2.228	1.114	47.0	40.8	46.2
Edge	1.0	3.693	3.693	149.9	144.4	144.6

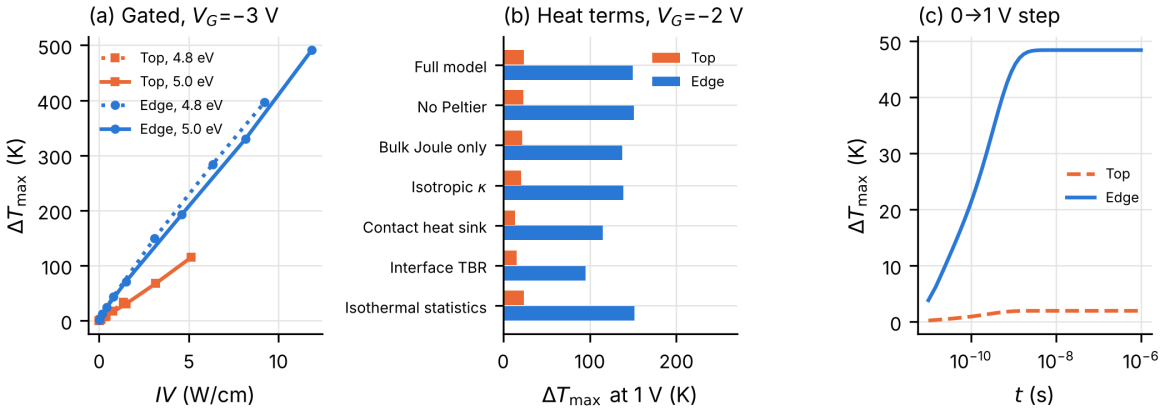


Figure 11: (a) Peak temperature rise versus dissipated power, gated on state. (b) Effect of each heat-model term at $V_D = 1$ V, $V_G = -2$ V, $\Phi_M = 5.0$ eV. (c) Response to a 0→1 V drain step at $V_G = -2$ V, simulated without tunnelling (steady-state rises of 48 K and 2 K); only the time constant is used in the text.

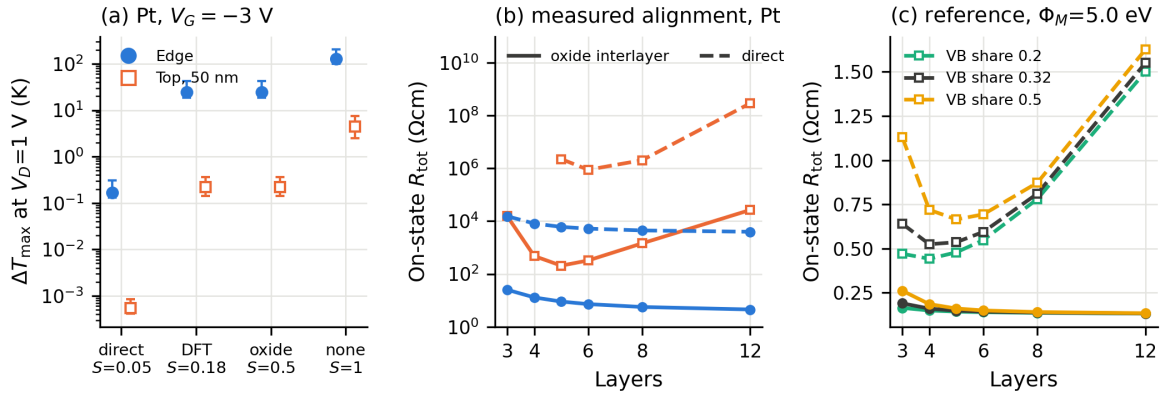


Figure 12: Measured alignment. (a) Peak temperature rise at $V_D = 1$ V, $V_G = -3$ V for Pt contacts and four pinning cases; bars span heat sinking through the metal (lower) and a three times larger R_{TBR} (upper). (b) On-state R_{tot} versus layer number for Pt with the oxide interlayer and direct pinning (missing top points did not converge). (c) Reference alignment ($\Phi_M = 5.0$ eV, $S = 1$) for valence-band shares of 0.2, 0.32 and 0.5. Filled: edge; open: top, 50 nm.

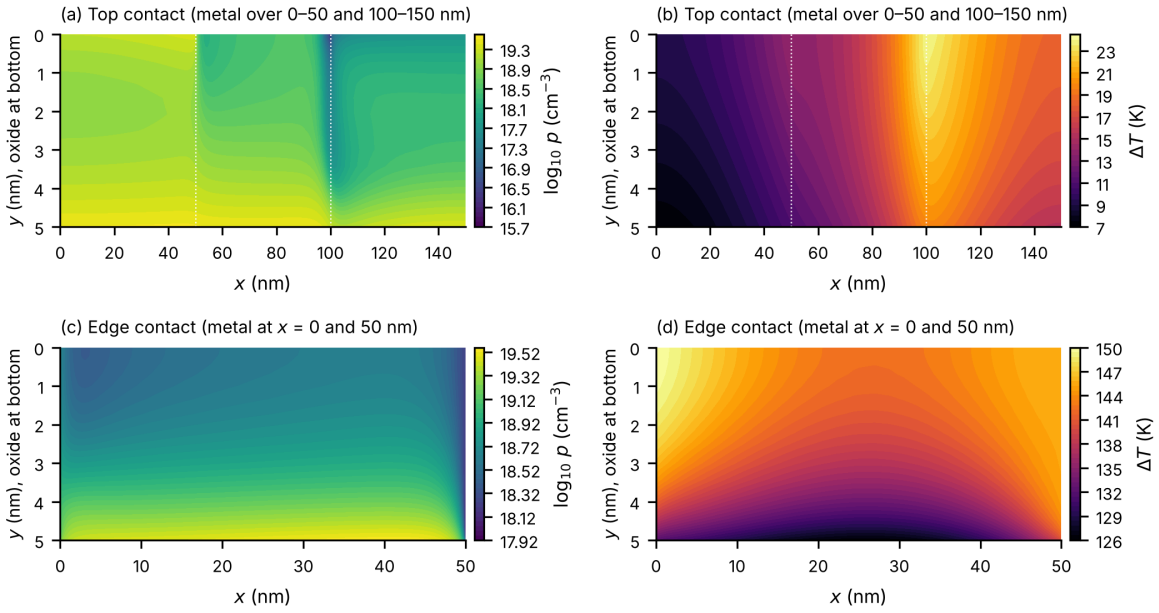


Figure 13: Hole density and temperature rise of gated devices at $V_D = 1$ V, $V_G = -2$ V, $\Phi_M = 5.0$ eV (full model). The vertical axis is stretched: the film is 5 nm thick. The oxide interface is at the bottom ($y = 5$ nm). Dotted lines in (a, b) mark the channel ends.

- a vertical access term $t/(2q\mu_\perp \bar{p})$;
- an under-contact sheet resistance $1/(q\mu_\parallel \bar{p} t)$;
- $R_c = \sqrt{\rho R_{\mathrm{sh}}}\coth(L_c/L_T)$.

Over 120 cases ($\phi_{Bp} = 0.05$–0.25 eV, $d_{\mathrm{vdW}} = 0.2$–0.4 nm, $L_c = 10$–100 nm, $r_\mu = 1$ and 0.01) the fully predictive version, with $\bar{p} = p_{\mathrm{eq}}$, has a median error of 11% and a 90th-percentile error of 33% (Fig. 14). Taking $\bar{p}$ from the simulation reduces the 90th percentile to 20%. The compact model is therefore adequate for circuit-level estimates when tunnelling is weak, that is, without strong gating.

3.10 Uncertainty quantification

We propagated the uncertainty of the least-known inputs by Monte Carlo sampling: 120 samples, each simulated for both geometries at 4.8 and 5.0 eV, ungated, at 0.05 V. The sampling ranges were:

- $\Delta E_V \sim \mathcal{N}(0, 0.1\ \mathrm{eV})$ and $S \sim U(0.3, 1)$, the same for both contacts;
- $d_{\mathrm{vdW}} \sim U(0.2, 0.4)$ nm and $\Phi_{\mathrm{vdW}} \sim U(1, 3)$ eV;
- log-uniform μ_p (20–100 cm^2/Vs), $\mu_\perp/\mu_\parallel$ (0.002–0.1) and N_A (10^{15}–10^{17} cm^{-3});
- $\varepsilon_r \sim U(5, 8)$ and $m_h \sim U(0.1, 0.3)$.

The resulting 5–95% range of R_{tot} spans seven to eight orders of magnitude: 61–$1.2 \times 10^9\ \Omega \cdot$ cm for the edge contact at 4.8 eV (Fig. 15a). The band alignment (ΔE_V, Spearman $\rho = -0.63$ to

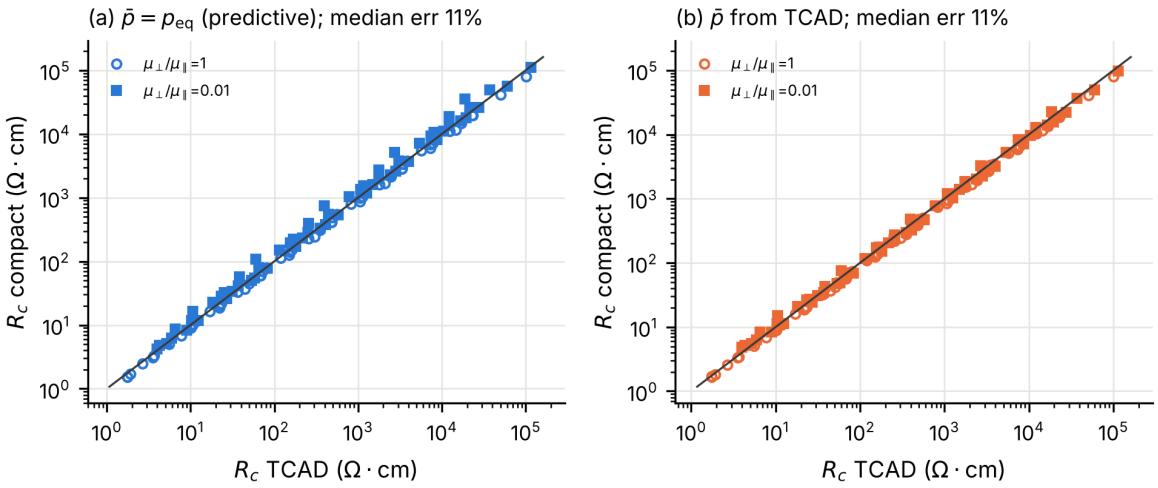


Figure 14: Compact transmission-line model against simulated top-contact R_c (no tunnelling), with (a) $\bar{p} = p_{\mathrm{eq}}$ and (b) $\bar{p}$ from the simulation.

-0.77) and the pinning factor ($\rho = -0.69$ to -0.81) account for nearly all of the variance. Their first-order indices are 0.47–0.77, while all other parameters stay at or near the estimator's noise floor of about 0.06–0.17 for $n = 120$ (Fig. 15b), so smaller contributions cannot be resolved. The *ranking* of the geometries is robust: when both contacts share the same pinning, the edge contact has lower resistance in every sample. The edge/top ratio has a median of 0.24, with a 5–95% range of 0.07–0.51 at 4.8 eV and 0.08–0.72 at 5.0 eV. Absolute resistances, by contrast, cannot be predicted to better than several orders of magnitude from these ranges alone. Section 3.5 narrows them using the measured pinning of InSe contacts; the remaining uncertainty is the absolute band alignment and the pinning of edge contacts.

This analysis is ungated, excludes the measured $S = 0.05$, and uses a spread of ΔE_V smaller than the difference between the reference, calculated and measured alignments (5.0, 5.2 and 5.9 eV). We therefore repeated it for the gated on state at the literature alignment (Pt, $V_G = -3$ V, $V_D = 0.05$ V; 240 samples; Fig. 16). The ranges were $\Delta E_V \sim \mathcal{N}(0, 0.3\ \mathrm{eV})$; log-uniform S from 0.05 to 0.5; a pinning level 0.1–0.6 eV below E_C; and the other ranges as above. Tunnelling carries essentially all of the current in every sample. The 5–95% range of R_{tot} is $10^{1.3}$–$10^{4.9}$ $\Omega \cdot$cm for the edge contact and $10^{2.7}$–$10^{7.3}$ $\Omega \cdot$cm for the top contact. The dominant inputs are now the pinning level (first-order index 0.29–0.39), the hole tunnelling mass (0.25–0.27) and S (0.18–0.22), against a noise floor of about 0.03. The band alignment falls to 0.06–0.07, because the pinning level is referenced to the band edges, as in the measurements. The vdW-gap parameters matter only for top contacts (Spearman $\rho \approx 0.2$). With shared pinning the edge contact has lower resistance in all 233 converged samples, with a median edge/top ratio of 0.0095 (5–95%: 0.0012–0.065). Seven top-contact runs did not converge, all at strong pinning and high barrier, where the top resistance is likely to be largest. When S_{edge} is sampled independently of S_{top}, the edge contact still has the lower resistance in 96% of samples. In the gated on state the geometric advantage is therefore robust to the pinning uncertainty that decides the ungated comparison of Sec. 3.4.

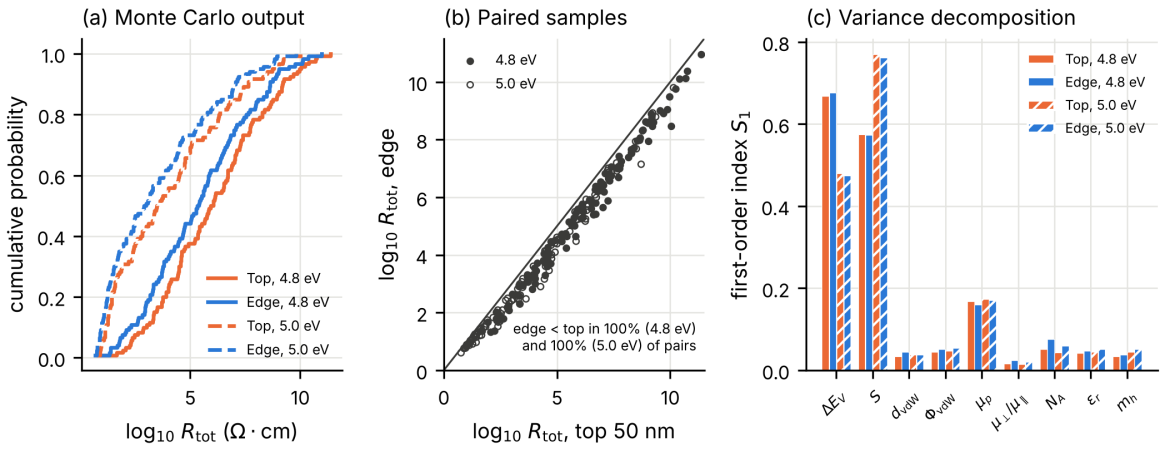


Figure 15: Monte Carlo propagation of parameter uncertainty (ungated, 0.05 V). (a) Cumulative distributions of $\log_{10} R_{\mathrm{tot}}$. (b) Paired samples: the same parameter draw simulated for the top (50 nm) and edge contact; points below the diagonal have lower edge resistance. (c) First-order variance indices (colour: geometry; hatching: 5.0 eV).

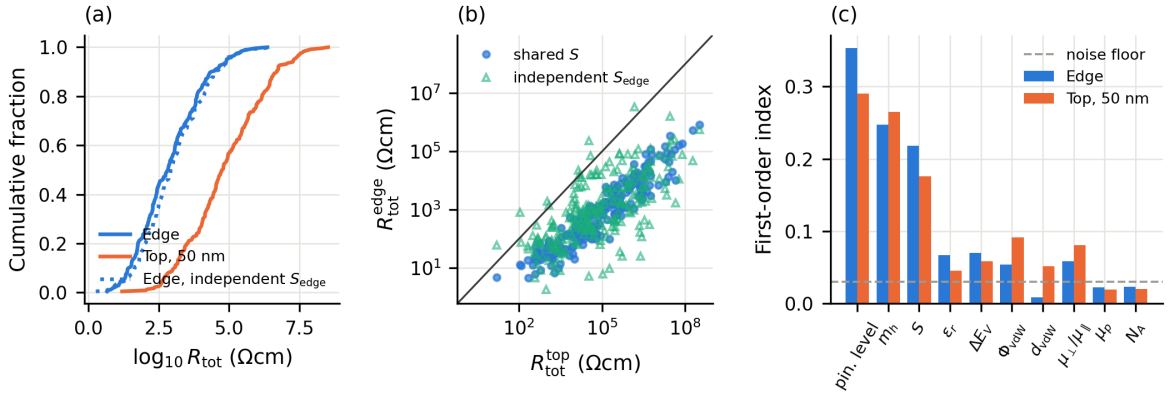


Figure 16: Monte Carlo analysis of the gated on state at the literature alignment (Pt, $V_G = -3$ V, $V_D = 0.05$ V, 240 samples). (a) Cumulative distributions of $\log_{10} R_{\mathrm{tot}}$. (b) Paired samples with shared S and with independently sampled S_{edge}; points below the diagonal have lower edge resistance. (c) Binned first-order indices with the permutation noise floor.

4 Conclusions

For hole injection into few-layer InSe p-FETs the model gives five design rules:

1. With the measured pinning of direct metal/InSe contacts ($S = 0.05$), every common metal gives a hole barrier of about 1.1 eV and an electron-injecting contact. Hole operation then relies entirely on gate-induced tunnelling, and the metal work function barely matters (2.4× from Ti to Pt).

2. The barrier must first be lowered by depinning: the measured oxide interlayer brings Pt to 0.52 eV and lowers the on-state resistance 700×.

3. For hole barriers up to 1 eV, edge contacts beat 50-nm top contacts in the on state, by 14× at 0.2 eV and 150× at 1 eV, because they avoid the vdW gap and the slow out-of-plane path. In ungated, barrier-limited operation this holds only if edge contacts are not more strongly pinned than top contacts; in the gated on state it held in 96% of samples with independently varied pinning. The pinning of edge contacts remains the interface property most worth measuring.

4. Nonlocal tunnelling is essential: thermionic and local enhancement models underestimate the on-current by orders of magnitude.

5. Top contacts have an optimum film thickness of four to six layers. Edge-contacted devices carry more current and need heat removal once the contacts are depinned.

The model is verified, benchmarked on Au/MoS_2 with one fitted parameter, anchored to the measured pinning of InSe contacts, and accompanied by a compact model and uncertainty bounds. Its main remaining limitations are the uncertain absolute band alignment of few-layer InSe and the unmeasured pinning of edge contacts. The vdW-gap transmission, the tunnelling mass and the interface thermal resistance are assumed, not measured. The one-dimensional WKB treatment is checked against quadrature on the same band profile (V6) but not against a two-dimensional quantum calculation. No InSe contact has been used as a benchmark. In the gated on state the hole tunnelling mass is, after the pinning level, the largest uncertainty, and on-state resistances at the measured pinning carry a mesh uncertainty of about 15%.

Statements and Declarations

Author contributions H.A. designed the study, developed the code, performed the simulations and analysis, and wrote the manuscript.

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Competing interests None.

Code and data availability https://github.com/Brukrish2006/InSe-Top-Edge-Contact-Resistace (`src_v3`, `data_v3`, `figures_v3`). The numbers quoted in Secs. 3.5 and 3.8 are regenerated from `data_v3` by `src_v3/check_numbers.py`. The gated Monte Carlo, measured-alignment thermal and thickness runs, mesh study and electron-injection check are produced by `src_v3/run_revision.py` and summarised by `src_v3/analyze_revision.py`.

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